SupplySetu (सप्लाईसेतु)

Technical Design Document (TDD)

AI-Powered B2B Ordering & Distribution Management Platform for FMCG Wholesalers

GitHub Repository: https://github.com/krishnayadav9793/SupplySetu

Live Staging URL: https://supplysetu.vercel.app

Client Territory: Jabalpur, Madhya Pradesh, India

Build Scope: 7-Day End-to-End Verified Core Implementation

System Status: Production-Ready (v1.0.0)

Automated Test Suite: 31/31 Tests Passing (100% Pass Rate)

1. Executive Summary & Problem Context

In India's FMCG wholesale distribution landscape, regional distributors connect major consumer packaged goods manufacturers (e.g., Parle, ITC, Hindustan Unilever, Britannia) with thousands of independent neighborhood retail stores (kiranas). The client for this project is a prominent packaged food and home-care wholesale distributor based in **Jabalpur, Madhya Pradesh**, serving approximately 1,800 kirana shops across commercial beats including Lordganj, Sarafa, and Ganjipura. The business employs 70 staff, operates 18 beat salesmen, and generates ≈ ₹85–100 Cr in annual turnover, processing over 600 orders daily.

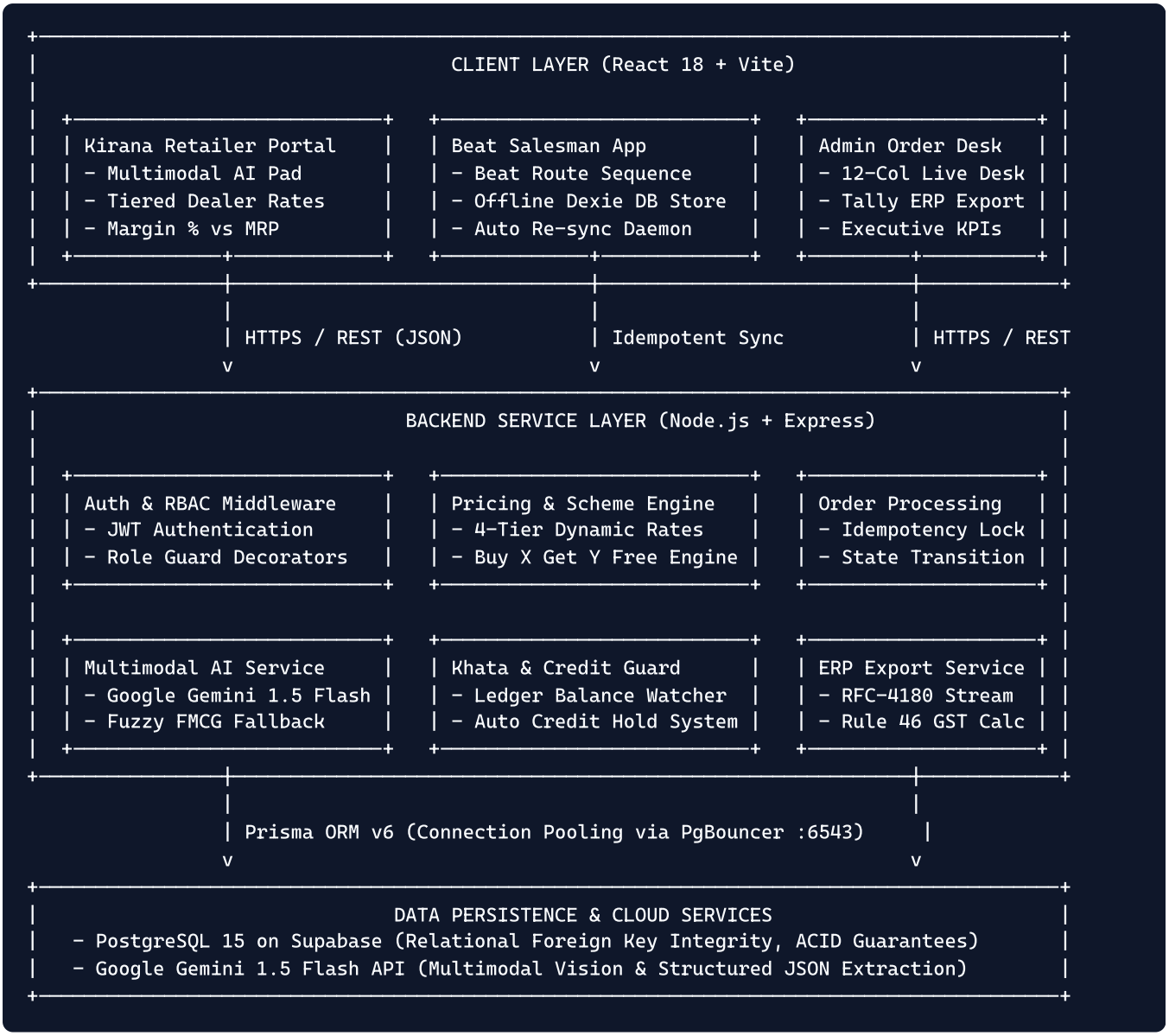
Despite this scale, the primary ordering pipeline relied entirely on manual, error-prone communication:

- **Unstructured WhatsApp Messages & Handwritten Slips:** Kiranas sent colloquial Hinglish text ("*bhaiya 5 peti parle-g chhota aur 2 peti surf excel 1 kilo*") or photos of handwritten slips (*parchi*).
- **Manual ERP Entry Bottleneck:** Six full-time office staff spent 4–5 hours every morning manually transcribing orders, looking up SKU numbers, and typing invoices into Tally ERP.
- **5% Order Error Rate:** Misidentified pack sizes and out-of-stock items caused frequent delivery returns, inventory discrepancies, and disputed khata ledger balances.
- **Connectivity Blackouts:** Beat salesmen visiting congested wholesale clusters encountered frequent cellular dead zones, disabling standard cloud-only web applications.

SupplySetu (सप्लाईसेतु) eliminates these failure points through a unified, multi-role platform featuring multimodal Gemini AI order ingestion, an offline-first Dexie IndexedDB sync engine for beat salesmen, real-time credit-limit controls, and one-click RFC-4180 Tally ERP CSV export.

2. System Architecture & Technical Tradeoffs

SupplySetu implements a decoupled client-server architecture engineered specifically for low-latency web interactions, offline field resilience, and reliable enterprise ERP export.


CLIENT LAYER (React 18 + Vite)
Kirana Retailer Portal
- Multimodal AI Pad
- Tiered Dealer Rates
- Margin % vs MRP
Beat Salesman App
- Beat Route Sequence
- Offline Dexie DB Store
- Auto Re-sync Daemon
Admin Order Desk
- 12-Col Live Desk
- Tally ERP Export
- Executive KPIs
HTTPS / REST (JSON)
Idempotent Sync
HTTPS / REST
BACKEND SERVICE LAYER (Node.js + Express)
Auth & RBAC Middleware
- JWT Authentication
- Role Guard Decorators
Pricing & Scheme Engine
- 4-Tier Dynamic Rates
- Buy X Get Y Free Engine
Order Processing
- Idempotency Lock
- State Transition
Multimodal AI Service
- Google Gemini 1.5 Flash
- Fuzzy FMCG Fallback
Khata & Credit Guard
- Ledger Balance Watcher
- Auto Credit Hold System
ERP Export Service
- RFC-4180 Stream
- Rule 46 GST Calc
Prisma ORM v6 (Connection Pooling via PgBouncer :6543)
DATA PERSISTENCE & CLOUD SERVICES
- PostgreSQL 15 on Supabase (Relational Foreign Key Integrity, ACID Guarantees)
- Google Gemini 1.5 Flash API (Multimodal Vision & Structured JSON Extraction)

Layer	Technology Chosen	Engineering Justification & Alternatives Evaluated
Frontend	React 18, Vite, Tailwind CSS	Fast build times and small bundle footprint (<650 kB gzipped). Tailwind CSS enables custom, high-contrast, touch-friendly UI components tailored for shopkeepers on budget smartphones without the overhead of heavy component libraries.
Offline Store	Dexie.js (IndexedDB wrapper)	LocalStorage is limited to 5MB and blocks the main UI thread. Dexie provides indexed, asynchronous, transactional client storage that allows salesmen to create and store complete order payloads offline in cellular dead zones.
Backend API	Node.js, Express, TypeScript	TypeScript guarantees shared contract interfaces between API routes and database models. The asynchronous Node event loop handles bursty batch exports and AI webhook requests with minimal overhead.
Database & ORM	PostgreSQL (Supabase), Prisma ORM v6	Financial khata ledgers, stock locks, and multi-tier discount hierarchies demand strict relational constraints and transaction rollback guarantees that NoSQL databases cannot reliably provide.
AI Ingestion	Google Gemini 1.5 Flash (`@google/genai`)	Low latency and native multimodal understanding. Gemini 1.5 Flash processes both noisy Hinglish text and photographed handwritten parchi slips in under 1.5 seconds at a fraction of the cost of legacy vision OCR pipelines.
ERP Integration	Streaming RFC-4180 CSV Engine	Direct Tally XML sockets require static IP mappings and desktop ODBC drivers that distributors resist installing. A RFC-4180 compliant CSV export integrates into Tally ERP's native Data Import function without client-side setup.

3. Data Model & Relational Schema

The relational data model is maintained in PostgreSQL via Prisma ORM v6, enforcing foreign-key cascade policies, unique constraints, and domain enums.

Entity	Primary Key	Key Attributes & Relationships
`User`	`id (UUID)`	`phone (Unique), passwordHash, name, role (ADMIN|SALESMAN|RETAILER), isActive`. Relates 1-to-1 with Retailer profile and 1-to-many with Orders placed as salesman.
`Retailer`	`id (UUID)`	`userId (FK), shopName, ownerName, city, creditLimit, outstandingBalance, priceTier (STANDARD|TIER_1|TIER_2|WHOLESALE), salesmanId (FK)`.
`Product`	`id (UUID)`	`skuCode (Unique), name, brand, category, packSize, mrp, basePrice, gstRate, isAvailable`. 1-to-1 with Inventory, 1-to-many with active Schemes and OrderItems.
`Inventory`	`id (UUID)`	`productId (FK, Unique), stockQuantity, minOrderQty, updatedAt`.
`Scheme`	`id (UUID)`	`productId (FK), schemeType (BUY_X_GET_Y_FREE|PERCENTAGE_DISCOUNT), minQuantity, freeQuantity, discountPercent, isActive`.
`Order`	`id (UUID)`	`orderNumber (Unique), retailerId (FK), salesmanId (FK), status (PENDING|APPROVED|REJECTED|DISPATCHED|DELIVERED), totalAmount, discountAmount, gstAmount, finalAmount, createdVia (PORTAL|AI_CAPTURE|SALESMAN_APP|OFFLINE_SYNC), isSynced`.
`OrderItem`	`id (UUID)`	`orderId (FK), productId (FK), quantity, unit, unitPrice, gstRate, discountAmount, freeQuantity, totalPrice`.

4. Core Business Logic Flows & Formulations

4.1 Flow 1: Dynamic Tiered Pricing & Retailer Margin Calculation

Wholesale distributors offer differentiated dealer rates based on kirana purchase volume and payment history. SupplySetu encodes this into four pricing tiers:

> $\text{Dealer Rate} = \text{Base Price} \times (1 - \text{Tier Discount Rate})$

> $\text{Retailer Margin \%} = ((\text{MRP} - \text{Dealer Rate}) / \text{MRP}) \times 100$

Pricing Tier	Discount Applied	Example: Parle-G 80g (MRP ₹10.00 / Box Base ₹597.80)	Retailer Margin vs MRP
`STANDARD`	0.0%	₹597.80 / box	**17.0%**
`TIER_1`	2.0%	₹585.84 / box	**18.6%**
`TIER_2`	4.0%	₹573.89 / box	**20.3%**
`WHOLESALE`	6.5%	₹558.94 / box	**22.4%**

4.2 Flow 2: Automated Promotional Scheme Engine

Schemes are evaluated at line-item aggregation time during checkout:

- **Buy X Get Y Free:** If order item quantity $Q \geq Q_{min}$, free bonus units are calculated deterministically:

  > $$Q_{free} = \lfloor Q / Q_{min} \rfloor \times Q_{bonus}$$

  Example: Parle-G (Buy 10 Get 1 Free). Ordering 25 boxes automatically adds $\lfloor 25/10 \rfloor \times 1 = 2$ free boxes without inflating the billable line item total.
- **Volume Percentage Discount:** If $Q \geq Q_{min}$, an additional trade discount is deducted from the line subtotal before GST computation:

  > $$\text{Discount Amount} = Q \times \text{Dealer Rate} \times (\text{Discount \%} / 100)$$

4.3 Flow 3: Credit Limit & Khata Control Circuit

To protect wholesale distributors from bad debt, every order passes automated credit verification:

> $$\text{Projected Balance} = \text{Outstanding Balance} + \text{New Order Final Amount}$$

- **Within Limit:** If *Projected Balance* ≤ *Credit Limit*, order proceeds to `PENDING` queue for standard fulfillment.
- **Credit Hold:** If *Projected Balance* > *Credit Limit*, order is flagged with `CREDIT_HOLD`. The retailer is notified with their available credit balance, and the Distributor Admin must explicitly approve the credit exception prior to warehouse picking.

4.4 Flow 4: Statutory GST Split (Rule 46 Compliance)

For distribution within Madhya Pradesh (State Code 23), intra-state invoices split GST equally between CGST and SGST:

> $$\text{CGST Amount} = \text{SGST Amount} = (\text{Taxable Subtotal} \times \text{GST Rate \%}) / 200$$

5. Multimodal AI Order Capture & Safety Guardrails

The AI ingestion pipeline converts unstructured natural-language Hinglish messages and photographed handwritten notebook slips (*parchi*) into verified, structured cart items.

> **Core Processing Steps:**
> 1. **Ingestion:** The user pastes Hinglish text or uploads an image of a handwritten *parchi* slip.
> 2. **Catalogue Grounding:** The request is dispatched to Google Gemini 1.5 Flash alongside the distributor's active product catalogue and schema constraints.
> 3. **Confidence Scoring:** The model extracts product candidates and returns structured JSON with item confidence scores (0.0 to 1.0).
> 4. **Anti-Hallucination Guardrail:** Items with confidence below 70% or SKU codes not present in the database are filtered out.
> 5. **Human-in-the-Loop Review:** The extracted items are presented in an interactive review modal where the user can adjust quantities, select pack sizes, and confirm prices.

Offline & Timeout Fallback: If the Gemini API call exceeds 8 seconds or experiences rate limits, the backend automatically falls back to an internal FMCG regex and fuzzy matching parser (`server/src/modules/ai/ai.service.ts`). This extracts numbers, packaging keywords (*peti, carton, darjan, kilo*), and brand tokens locally without failing the user's order flow.

6. Offline-First Beat Salesman Engine

Salesmen in congested market sectors (Lordganj, Sarafa) regularly lose cellular connectivity. SupplySetu implements a reliable offline architecture using **Dexie.js (IndexedDB)**:

- **Local Catalogue Caching:** Product and retailer lists are cached locally in browser IndexedDB.
- **Idempotency Key Generation:** Each offline order receives a cryptographically generated UUID v4 (`clientOrderId`) at the time of creation.
- **Transactional Store:** Orders are written to Dexie's `offlineOrders` store with status `PENDING_SYNC`.
- **Automatic Synchronization:** When connectivity is restored, a background listener submits queued orders to `POST /api/orders/sync`. The server checks the `clientOrderId` to prevent duplicate insertions, updates local records to `SYNCED`, and refreshes the salesman's order log.

7. UI & Design Decisions: Key Screens & UX Flow

- **Retailer Dashboard (`RetailerDashboard.tsx`):** Mobile-first catalogue with live stock indicators, transparent retailer profit margin badges (e.g., "*Your Margin: 18.6%*"), 1-click reorder functionality, and a real-time khata balance meter.
- **AI Order Pad Modal (`AiOrderPadModal.tsx`):** Split-view ingestion interface supporting text paste, speech dictation, or handwritten parchi image upload, followed by an interactive item verification table

with confidence scores.

- **Beat Salesman Pad (`SalesmanDashboard.tsx`):** Grouped by daily beat route sequences, displaying outstanding kirana balances before shop visits, quick order punch buttons, and an offline synchronization status badge.
- **Distributor Admin Order Desk (`AdminDashboard.tsx`):** 12-column operational table with credit hold warnings, batch approval actions, streaming RFC-4180 Tally ERP CSV export, and an Executive Analytics suite featuring a 14-day kirana churn detection radar with one-click WhatsApp re-engagement links.

8. Scope Boundaries: Deliberate Cuts & Next Steps

Feature Deliberately Cut	Technical & Business Justification	Chosen 7-Day Production Alternative
Native Android APK	Building and distributing native binaries across fragmented Android devices introduces installation friction for non-technical shopkeepers.	Responsive Progressive Web App (PWA) with Dexie IndexedDB offline support that works instantly in any mobile browser without installation.
Direct Tally XML Socket Daemon	Direct bidirectional XML sockets require static IP configuration, port forwarding, and background services on the distributor's office computer.	Streaming RFC-4180 compliant CSV export engine matching Tally ERP's native import template, allowing immediate adoption with zero setup.
Instant Payment Gateway Checkout	Wholesale FMCG operates on rolling 7 to 15-day credit terms (*khata*). Immediate UPI/card payment checkout would introduce adoption barriers.	Implemented a dedicated Khata Ledger management system with credit ceilings, payment collection logging, and balance tracking.

Post-Submission Roadmap: 1) Official WhatsApp Cloud API webhook receiver for automated chat ordering; 2) GeoJSON salesman beat route optimization; 3) Predictive kirana replenishment demand models; 4) Background Tally ODBC sync daemon.

9. System Assumptions

1. **State GST Jurisdiction:** Operations are localized to Jabalpur, Madhya Pradesh (State Code 23), with standard 50/50 CGST/SGST splits on intra-state sales.
2. **Wholesale Packaging:** Products are sold in commercial packaging units (`BOX` , `CARTON` , `BAG`). Loose individual pieces are not handled.
3. **Credit Terms:** Retailers operate on rolling 15-day settlement cycles with credit ceilings enforced prior to order dispatch.
4. **Single-Distributor Tenancy:** Designed for a single distributor entity serving multiple retail accounts across assigned geographical beats.
5. **Simulated OTP Sign-in:** DLT registration delays are avoided by simulating mobile OTP verification with pre-configured demo credentials.

10. Automated Verification Suite (31 Tests, 100% Pass Rate)

All business logic, pricing calculations, scheme mechanics, credit controls, offline sync idempotency, and Tally export compliance are validated via an automated test suite (`npm --prefix server test`):

Category	Tests	Key Verification Logic	Status
Pricing Tiers	5	Standard, Tier 1 (2%), Tier 2 (4%), Wholesale (6.5%), MRP margin %	✓ Passed
Scheme Engine	4	Buy 10 Get 1 Free thresholds, multiple brackets, volume discounts	✓ Passed
Credit Controls	2	Clean approval under limit vs automated Credit Hold flag	✓ Passed
Offline Sync	2	UUID clientOrderId idempotency and duplicate rejection	✓ Passed
Tally ERP Export	2	RFC-4180 CSV escaping for store names and ERP header validation	✓ Passed
Multimodal AI	3	Parchi image processing, fallback regex parsing, non-catalogue rejection	✓ Passed
Khata Ledger	2	Credit utilization calculation and balance updates upon payment	✓ Passed
B2B Tax Invoice	2	Rule 46 CGST/SGST 50/50 split and HSN chapter classifications	✓ Passed
Order Lifecycle	2	State transitions (PENDING → APPROVED → DELIVERED) and repeat orders	✓ Passed
Executive KPIs	2	GMV, AOV, fulfillment rate, and brand revenue distribution	✓ Passed
Churn Radar	2	14-day inactivity threshold and WhatsApp re-engagement URL formatting	✓ Passed
Financial Risk	2	Receivables aging brackets (0-15d, 16-30d, 30+d) and utilization warnings	✓ Passed
Sales Performance	1	Daily beat quota attainment and spot commissions	✓ Passed
Overall	**31**	**All End-to-End Business Logic Invariants Validated**	**✓ 31/31 (100%)**